

Capacity-Gated Projection Dynamics:

A Typed Hybrid Stochastic Model and Executable Specification

Peter Nero

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Abstract

This paper turns the qualitative idea of capacity-gated projection into a complete model specification. The model contains finitely many particles, declared relational summary fields, normalized admissibility rows, a bottleneck reserve, within-mode drift and diffusion, flow and jump sets, and a deterministic reset map or Markov reset kernel. These data define a hybrid stochastic particle system. They are not consequences of a capacity scalar alone.

The distinction repairs several ambiguities in the previous version. The logarithmic expression used there is a barrier potential, and its negative gradient is a genuine feedback force or penalty within the reduced model. A capacity-dependent speed cap and noise schedule are constitutive choices. Reaching a guard does not select a successor state until a reset law is supplied. Shared relational fields generally produce classical collective dependence, not quantum entanglement. The use of a Gaussian kernel also creates all-to-all mean-field coupling rather than strictly local interaction.

We give a typed model record, a smooth rowwise barrier, a conditional well-posedness theorem up to guard accumulation, an explicit no-selection result for an unspecified reset, a generator criterion for classical nonfactorization, and an event-driven Euler–Maruyama specification that does not silently step through the guard. A two-particle circle model shows every object in closed form, including a translation-covariant reset kernel. A parameter and provenance ledger separates quantities inherited from an MTT admissibility problem from modeling choices and numerical controls. Boundary layers, hesitation, channels, sprouting, and persistent basins are therefore simulation observables to test under stated parameters, not inevitable consequences of projection. The result is a reproducible conditional model and a clear contract for promoting it to a physical application.

Revision note: Version 3

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Reason Version 2 presented a logarithmic barrier and its gradient as though they were not a force or penalty, left the response at zero capacity undefined, and described regime changes, irreversibility, nonfactorization, boundary layers, state selection, and absence of close analogues as necessary consequences. The displayed update is instead a hybrid constrained stochastic particle model. Its continuation requires flow domains, guards, reset laws, invariance conditions, and a solution concept.

Resolution The capacity input is now the fixed normalized row record developed in the preceding capacity papers. The revision declares every additional dynamical ingredient, uses a rowwise barrier on a positive flow domain, defines deterministic and stochastic resets, states regularity and nonaccumulation hypotheses, and distinguishes classical shared-field dependence from quantum entanglement.

Retained content

The useful construction survives: coarse relational summaries can be fed back into particle dynamics through a weakest-row admissibility gate, and a guard can trigger a change of effective chart or mode.

Remaining boundary

MTT does not currently select a universal relational kernel, force law, noise schedule, guard threshold, or reset kernel. A physical application must derive those data from its upper operator or action and validate the result against observations not used in construction.

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1 Question, answer, and scope

The question of this paper is deliberately concrete:

What must be specified before “capacity-gated projection dynamics” is an actual dynamical system rather than a suggestive phrase?

The answer is more structured than a single update equation. A static capacity record tells us where a chosen effective description is certified. It does not say how a state moves, whether a boundary is invariant, what happens at first exit, or how randomness enters. The corrected MTT capacity chain makes this separation explicit:

1. the MTT Foundation owns the admissibility ledger and the stop/continue/reset alternatives [13];
2. the normalized-margin paper owns signed slacks, scales, reserve rows, bottleneck capacity, and metric clearance [11];
3. the control-budget paper owns perturbation cost and cumulative consumption certificates [10]; and
4. the transport paper explains which additional data promote a static reserve to a continuous-time model [12].

This paper supplies one such promotion for an interacting particle model. It does not claim that the promotion is unique or selected by present MTT geometry. Its contribution is a typed construction whose assumptions can be inspected, implemented, varied, and falsified.

The resulting mathematical neighbors are familiar: hybrid systems use flow sets, jump sets, flow maps, and jump maps [5, 6]; stochastic hybrid systems add diffusion and reset kernels [9]; viability theory studies evolution constrained to a set [2]; barrier functions enforce forward-invariance conditions [1]; and mean-field particle systems couple particles through aggregate empirical fields [4]. The MTT-specific feature is not a new existence theory for these objects. It is the provenance-aware use of a multirow admissibility record as the gate and the refusal to confuse guard hitting with a sourced physical outcome.

Object	Inherited from a capacity problem	Supplied by this reduced model
State and effective chart	Declared domain and admissibility rows	Particle coordinates, velocities, and mode labels
Normalization	Row tolerances and positive scales	None; the inherited normalization is frozen
Relational summary	Only if derived by the application	Kernels, weights, field evaluation, and sampling rule
Within-mode evolution	Only if an upper action or operator supplies it	Drift, damping, diffusion, speed gate, and barrier gain
First-exit logic	Stop/continue/reset alternatives	Numerical guard threshold and event detector
Post-exit continuation	Not selected by a scalar reserve	Stop rule, deterministic reset, or Markov reset kernel
Physical interpretation	Only at the tier proved by the source theory	No automatic gravity, quantum, cognitive, or thermodynamic meaning

Table 1: Ownership of the ingredients. Every right-column entry is constitutive or numerical unless a later source theorem derives it.

2 What is inherited and what is supplied

The easiest way to prevent hidden assumptions is to divide the model into two layers.

This division changes the status of the algorithm. It is a *conditional reconstruction*: if a particular application provides the rows and accepts the declared constitutive data, the hybrid process is well defined under the hypotheses below. It is not a derivation of those data from projection alone.

3 Particle state, relational summaries, and reserve rows

3.1 Microscopic state

Let Ω be a smooth compact d -dimensional manifold. A flat torus is a convenient boundary-free simulation domain; a bounded Euclidean domain may instead be used together with a declared reflecting, absorbing, or periodic boundary rule. For N labeled particles, write

$$X = (x_1, \dots, x_N) \in \Omega^N, \quad V = (v_1, \dots, v_N) \in (T\Omega)^N.$$

Let q lie in a finite mode set $\mathcal{Q}$. The reduced state is

$$z = (X, V, q) \in \mathcal{E}.$$

Extra memory, records, or internal states must be appended to $\mathcal{E}$ if an application uses them. They are not implicit.

3.2 Relational field map

Choose K summary channels. Channel k has a kernel $K_k : \Omega \times \Omega \rightarrow \mathbb{R}$ and fixed weights w_{kj} . Define

$$f_k(x; X) = \sum_{j=1}^N w_{kj} K_k(x, x_j).$$

The map

$$\mathcal{P} : X \mapsto (f_1(\cdot; X), \dots, f_K(\cdot; X))$$

is the relational summary map. It is an explicit coarse observable, not a new physical substance.

Compactly supported K_k give finite-range coupling. Gaussian kernels are nonzero at every finite separation and therefore give all-to-all coupling, even though their influence decays with distance. Calling the latter update “strictly local” would be incorrect.

Proposition 3.1 (A sufficient source of noninjectivity). *Suppose particles $a \neq b$ have equal weights in every channel: $w_{ka} = w_{kb}$ for all k . If labels belong to the microscopic state, then exchanging x_a and x_b leaves $\mathcal{P}(X)$ unchanged. Hence $\mathcal{P}$ is noninjective on labeled configurations whenever the exchange changes X .*

Proof. Every f_k is a sum. The two exchanged summands have equal coefficients, so their sum and therefore every field f_k is unchanged. $\square$

Remark 3.2. The proposition proves only a model-relative loss of label information. If particles are physically indistinguishable and the microscopic state is already quotiented by permutations, this argument does not prove further noninjectivity. Nor does noninjectivity alone imply irreversibility; a section may exist on the selected range.

3.3 Signed normalized reserve rows

For each particle i , mode q , and constraint row $a \in \{1, \dots, m\}$, let

$$\sigma_{ia}^q(X, V)$$

be a signed slack and let $s_{ia}^q > 0$ have the same units. Define

$$r_{ia}^q(z) = \frac{\sigma_{ia}^q(X, V)}{s_{ia}^q}, \quad c_q(z) = \min_{i,a} r_{ia}^q(z).$$

The vector $r^q = (r_{ia}^q)$ is primary. The scalar c_q is its bottleneck compression. No row is clipped at zero: negative values retain which condition failed and by how much.

The exact admissible set in mode q is

$$\mathcal{A}_q = \{z : r_{ia}^q(z) > 0 \text{ for every } i, a\}.$$

The active set is

$$I_q(z) = \{(i, a) : r_{ia}^q(z) = c_q(z)\}.$$

The minimum is generally nonsmooth when $I_q(z)$ contains more than one row. This is why the smooth barrier below is built row by row rather than by differentiating the bottleneck without qualification.

4 The typed hybrid record

Definition 4.1 (Capacity-gated hybrid record). A capacity-gated hybrid record is the tuple

$$\mathfrak{H} = (\mathcal{E}, \mathcal{Q}, \mathcal{P}, r, \eta, \mathcal{C}, \mathcal{D}, b, \Sigma, U, \mathcal{R}, \mathcal{B}, \mathfrak{s}, \mathfrak{p}),$$

where:

1. $\mathcal{E}$ is the state space and $\mathcal{Q}$ the finite mode set;
2. $\mathcal{P}$ is the relational summary map;
3. $r = (r_{ia}^q)$ is one fixed normalized reserve record;
4. $\eta = (\eta_q)$ is a nonnegative numerical guard margin;
5. $\mathcal{C}_q$ is the flow set and $\mathcal{D}_q$ the jump or stop set;
6. b_q and Σ_q are within-mode drift and diffusion;
7. U_q is a declared barrier or feedback potential;
8. $\mathcal{R}_q(z, dz')$ is a deterministic or Markov reset law;
9. $\mathcal{B}$ is the spatial boundary rule;
10. $\mathfrak{s}$ is the strong, weak, martingale, or numerical solution concept; and
11. $\mathfrak{p}$ records the source, units, version, and hash of every nontrivial ingredient.

The exact failure boundary is $c_q = 0$. Numerical simulation normally jumps earlier. For a chosen $\eta_q > 0$, define

$$\mathcal{C}_q = \{z : c_q(z) \geq \eta_q\}, \quad \mathcal{D}_q = \{z : c_q(z) \leq \eta_q\}.$$

Their overlap at equality permits event handling. One may instead use disjoint conventions, hysteresis, or a set-valued rule, but the convention must be stated.

Remark 4.2 (Guard margin versus physics). The value η_q is a numerical or control margin unless a source theorem assigns it physical meaning. A result that changes when η_q changes is a property of the implemented guard, not of the exact boundary alone.

5 Within-mode dynamics and the barrier force

5.1 A smooth rowwise barrier

Choose nonnegative barrier weights α_{ia}^q . On the exact admissible set define

$$U_q(z) = - \sum_{i,a} \alpha_{ia}^q \log r_{ia}^q(z).$$

On any closed numerical flow set $c_q \geq \eta_q > 0$, this function is smooth whenever the rows are smooth. Its gradient is

$$\nabla U_q(z) = - \sum_{i,a} \frac{\alpha_{ia}^q}{r_{ia}^q(z)} \nabla r_{ia}^q(z).$$

Unlike $-\log(\varepsilon + c_q)$, this expression does not hide active-row ties inside an unjustified ordinary gradient.

Proposition 5.1 (The barrier is dynamical feedback). *If the reduced equation contains $-\nabla U_q$, then U_q is a potential and $-\nabla U_q$ is a force or feedback term in that reduced equation. This remains true if U_q was motivated by representational admissibility rather than by microscopic energy.*

Proof. A force term is identified by its role in the evolution equation, not by its interpretation. The displayed vector is the negative gradient of a scalar potential and changes the acceleration or drift. It is therefore a gradient feedback force in the reduced model. No claim that it equals a fundamental physical force follows. $\square$

The logarithmic barrier is familiar from constrained optimization and control. Control barrier functions can certify forward invariance when their derivative inequalities are verified [1]. Merely placing a logarithm in an update does not prove invariance. That requires a tangency, viability, or barrier inequality for the complete drift and diffusion.

5.2 Stochastic flow

Within mode q , a transparent second-order model is

$$dX_t = V_t dt, \tag{1}$$

$$dV_t = [B_q(X_t, V_t) - \Gamma_q(X_t, V_t)V_t - G_q(X_t, V_t)\nabla_X U_q(X_t, V_t)]dt + \Sigma_q(X_t, V_t)dW_t, \tag{2}$$

until the first hitting time of $\mathcal{D}_q$. Here B_q is a declared base drift, Γ_q is damping, G_q is a mobility or gain, and W_t is a Brownian driver of declared dimension and covariance convention.

Every term is optional but none is implicit. Setting $\Sigma_q = 0$ gives a deterministic hybrid system. A first-order model can omit V . A discrete-time model can be specified directly, but must still define what happens when a step crosses the guard.

5.3 Speed and noise gates

A speed gate can be imposed through a projection

$$\Pi_{V_q(c)}(v) = \begin{cases} v, & \|v\| \leq V_q(c), \\ V_q(c) v / \|v\|, & \|v\| > V_q(c), \end{cases}$$

or through a drift that enforces an invariant velocity set. A diffusion schedule may use $\Sigma_q(z) = \chi_q(c_q(z))\Sigma_q^0(z)$.

These choices can produce slow motion or reduced noise near a guard, but that behavior has been inserted through V_q or χ_q . It is not a theorem about finite capacity. Conversely, a model with constant diffusion and no speed gate need not “hesitate” at all.

6 Guards, resets, and continuation

Let

$$\tau_1 = \inf\{t > 0 : Z_t \in \mathcal{D}_{q_0}\}$$

be the first guard-hitting time. At τ_1 there are four distinct contracts.

1. **Stop.** The effective trajectory terminates at τ_1 .
2. **Continue in another chart.** A declared transition map changes coordinates while representing the same upper state.
3. **Deterministic reset.** A measurable map $R_q : \mathcal{D}_q \rightarrow \mathcal{E}$ sets $Z_{\tau_1} = R_q(Z_{\tau_1}^-)$.

4. **Stochastic reset.** A Markov kernel $\mathcal{R}_q(z, dz')$ samples the post-jump state.

Boundary-hitting diffusions with deterministic resets and their Fokker–Planck equations are established mathematical objects [3]. The reset is part of the model data, not a consequence of the boundary.

Definition 6.1 (Reset support contract). Choose a restart margin $\eta_q^{\text{reset}} > \eta_q$. A reset law is admissibility restoring if

$$\mathcal{R}_q(z, \{z' : c_{q'}(z') \geq \eta_{q'}^{\text{reset}}\}) = 1$$

for every guard state z at which a reset is permitted, with q' the post-jump mode contained in z' .

Proposition 6.2 (A guard does not select an outcome). *Suppose a guard state z admits two distinct admissibility-restoring restart states. Then the state space, flow law, reserve rows, and guard do not determine a unique post-guard trajectory from z .*

Proof. Let $z_1 \neq z_2$ be the two restart states. The deterministic kernels δ_{z_1} and δ_{z_2} have the same state space, flow law, rows, and guard but produce different continuations. A nontrivial convex mixture of these kernels produces a third continuation law. Therefore the pre-guard data do not select one of them. $\square$

Corollary 6.3 (No automatic irreversibility). *Guard hitting alone proves neither reversibility nor irreversibility. Those properties depend on the continuation rule, retained records, coarse-graining, and the state space on which reversal is tested.*

A bijective chart transition may preserve information. A many-to-one reset may discard it. A stochastic reset may introduce an outcome law. Those are different mechanisms and should not be merged under the word “selection.”

7 Conditional well-posedness

The next result states exactly what the model obtains from standard stochastic and hybrid-system theory.

Theorem 7.1 (Guarded hybrid existence and uniqueness). *Assume for every mode q that:*

1. Ω is compact, or the coefficients satisfy standard nonexplosion growth bounds;
2. every reserve row is C^2 on an open neighborhood of $\mathcal{C}_q$ and $\eta_q > 0$;
3. the drift and diffusion in Eqs. (1) and (2) are locally Lipschitz on that neighborhood;
4. the spatial boundary rule is well posed;
5. $\mathcal{R}_q$ is a measurable probability kernel satisfying Theorem 6.1; and
6. almost surely, jump times have no finite accumulation point.

Then an initial state in a flow or restart set determines a unique strong within-mode solution up to its next guard time. Concatenating that solution with the reset kernels defines a strong Markov hybrid process for all finite times. If the reset is deterministic, the same statement holds with a deterministic jump map.

Proof. Because $c_q \geq \eta_q > 0$ on the closed numerical flow set, every denominator in ∇U_q is bounded away from zero. The assumed row regularity makes the barrier gradient locally Lipschitz there. Standard SDE existence and pathwise uniqueness therefore apply until the first exit from the flow domain. At the exit, the measurable reset kernel supplies the conditional law of the next initial state. Iterating this construction gives the usual piecing-out construction of a stochastic hybrid process [9]. The no-accumulation hypothesis ensures that only finitely many pieces occur on each finite time interval. $\square$

Remark 7.2 (What the theorem does not prove). The theorem does not prove that a guard is reached, that a reset state is physically selected, that the process converges, that the safe set is invariant, or that the model is a faithful description of a target system. Those are separate reachability, source, stability, and validation questions.

Remark 7.3 (Zeno behavior). A restart margin alone does not always rule out infinitely many jumps in finite time. One may establish a positive dwell time from bounded drift and a positive reserve gap, introduce hysteresis or a timer, or treat Zeno completion explicitly. The chosen route belongs in the model record.

8 Classical collective dependence

The shared fields make particle updates dependent. This is important, but it should be named correctly.

Let $\mathcal{L}$ be the within-mode generator. If particles were dynamically independent, it would decompose as

$$\mathcal{L} = \sum_{i=1}^N \mathcal{L}_i,$$

where $\mathcal{L}_i$ acts only on the coordinates of particle i . Shared-field feedback generally violates this decomposition.

Proposition 8.1 (Cross-dependence criterion). *Suppose the drift of particle i contains $\nabla_{x_i} f_k(x_i; X)$ and for some $j \neq i$*

$$w_{kj} \nabla_{x_i} K_k(x_i, x_j) \neq 0$$

on an open set. Then the drift of particle i depends on x_j on that set. The generator is not a sum of single-particle generators there, and the finite-time transition law is not generically a product of independent single-particle transition laws.

Proof. Differentiating the field gives

$$\nabla_{x_i} f_k(x_i; X) = \sum_{\ell=1}^N w_{k\ell} \nabla_1 K_k(x_i, x_\ell).$$

The nonzero j term changes when x_j changes while x_i is held fixed. Thus the i drift cannot be a function of the i state alone. The generator consequently contains cross-coordinate dependence, which generically prevents product transition kernels. $\square$

This is classical collective correlation or mean-field dependence. A claim of quantum entanglement would additionally require a quantum state space or observable algebra, a subsystem composition rule, a state, and a separability or entanglement criterion. Those data are absent here. Quantum separability is a precise property of states on composite quantum systems [15, 8]; classical nonfactorization of a particle transition law is not a substitute for it.

9 An executable event-driven specification

The continuous model can be simulated by an event-aware discretization. The following specification is intentionally explicit about failure.

INPUT:

```
model record H, final time T, initial state z0,  
step h, event tolerance tol, random seed
```

REQUIRE:

```
z0 is in a flow or restart set  
all source hashes and units in the provenance ledger match
```

```
t <- 0  
z <- z0  
while t < T:  
  q <- mode(z)  
  compute every signed row  $r_{ia}^q(z)$   
  if  $\min(r_{ia}^q(z)) \leq \eta_q$ :  
    if policy is STOP:  
      return trajectory with certified first-exit record  
    z <- sample  $R_q(z, \cdot)$  or evaluate deterministic reset  $R_q(z)$   
    verify reset support and record random draw  
    continue  
  
  compute relational fields and their derivatives  
  compute rowwise barrier gradient  
  propose one Euler--Maruyama step  $z_{\text{trial}}$   
  
  if  $z_{\text{trial}}$  stays in the flow set:  
    accept  $z_{\text{trial}}$   
    t <- t + h  
  else:  
    bracket the first guard crossing on the same noise increment  
    refine the crossing time to tolerance tol  
    advance to the guard and apply the declared stop/reset policy
```

OUTPUT:

```
path, mode history, row history, active rows,  
guard events, reset draws, seed, parameter ledger,  
source hashes, tolerances
```

9.1 Why naive clipping is not enough

Replacing every negative row by zero before event detection loses the failed-row identity and violation size. Accepting an Euler step that jumps from $c_q > \eta_q$ to $c_q < 0$ also invents dynamics outside the declared flow domain. The event-aware scheme brackets that crossing and invokes the declared continuation rule.

9.2 Numerical versus exact statements

Euler–Maruyama is an approximation, not the hybrid process itself. Convergence requires the usual coefficient and event-detection hypotheses; hybrid simulators need additional care near jumps [14, 7]. A reproducibility packet should report step refinement, event tolerance refinement, random seeds, and the fraction of trajectories affected by guard localization.

10 Complete parameter and provenance ledger

The phrase “no hidden parameters” is justified only by publishing all of them. A minimal ledger contains the following rows.

Parameters may be fixed for a toy model without being fitted. That makes the example reproducible, not parameter free. For a physical claim, the ledger must also distinguish:

- values selected by an upstream theorem;
- conventional choices that change coordinates but not predictions;
- numerical controls removed by convergence;
- genuine constitutive parameters;
- quantities fitted to the same data later reported; and
- held-out observables used only for validation.

11 A closed-form two-particle circle model

This example is not a physical fit. Its purpose is to show that every item in the hybrid record can be filled without ambiguity.

Let $\Omega = \mathbb{R}/L\mathbb{Z}$ and $N = 2$. Use the periodic kernel

$$K(y) = 1 + \cos\left(\frac{2\pi y}{L}\right).$$

Define one relational field

$$f(x; X) = K(x - x_1) + K(x - x_2)$$

and one reserve row per particle,

$$r_i(X) = 4 - f(x_i; X).$$

At coincidence $x_1 = x_2$, each row is zero because $f(x_i; X) = 4$. At antipodal separation $x_2 = x_1 + L/2$, each row equals 2 because the self-term is 2 and the cross-term is zero.

Choose a guard $\eta \in (0, 2)$ and barrier

$$U(X) = -\alpha(\log r_1(X) + \log r_2(X))$$

on $r_1, r_2 \geq \eta$. Let the within-mode dynamics be

$$dx_i = v_i dt, \quad dv_i = [-\gamma v_i - \partial_{x_i} U(X)] dt + \sigma dW_i.$$

All four parameters $L, \alpha, \gamma, \sigma$ and the guard η are declared model inputs.

Entry	Mathematical role	Required provenance
N, d, Ω	particle count, dimension, domain	physical source, model choice, or test-grid declaration
Boundary rule	periodic, reflecting, absorbing, or reset	domain theorem or constitutive choice
K_k, σ_k	relational kernels and ranges	upper overlap calculation or chosen interaction scale
w_{kj}	channel weights	symmetry/source rule or explicit model parameter
σ_{ia}^q	signed admissibility slacks	exact source condition and convention
s_{ia}^q	row normalization scales	units, tolerance, and calibration source
η_q	numerical guard margin	numerical choice or sourced physical threshold
α_{ia}^q	barrier gains	control design or source derivation
B_q, Γ_q, G_q	base drift, damping, mobility	action/operator derivation or constitutive model
Σ_q	diffusion and covariance	noise source, units, and stochastic convention
V_q, χ_q	speed and noise gates	constitutive schedule and regularity
$\mathcal{R}_q$	deterministic or stochastic continuation	source rule, support proof, and randomization semantics
h, tol	time step and event tolerance	convergence study
Seed and generator	reproducible random stream	algorithm, version, and seed
Initial law	starting ensemble	independent protocol, not selected after seeing outcomes
Diagnostics	reported observables and estimators	definitions fixed before interpretation

Table 2: Minimal parameter and provenance ledger. A row is not derived merely because it has been given an MTT name.

At the guard, choose a random global phase ξ uniformly on the circle and reset

$$(x_1, x_2, v_1, v_2) \mapsto (\xi, \xi + L/2, 0, 0).$$

This Markov reset is translation covariant and returns both rows to 2. It therefore satisfies the reset support contract for every $\eta^{\text{reset}} < 2$.

The same flow and guard could instead stop, reverse the relative velocity, or reset to another safe configuration. Theorem 6.2 explains why none of these outcomes follows from the guard itself. The example also shows classical dependence: the force on particle 1 depends on $x_2 - x_1$.

12 Behavioral diagnostics, not automatic conclusions

The previous version described several behaviors as necessary. In the typed model they become measurable questions.

Label	Operational diagnostic	What must be shown
Boundary layer	systematic change of drift, residence time, or density as $c_q \downarrow \eta_q$	asymptotic analysis or step-refined simulation, separated from a barrier inserted by design
Hesitation	reduced speed or increased dwell time near the guard	dependence on the declared speed, damping, and noise schedules
Channel	persistent anisotropic path concentration	geometry of the reserve rows and comparison with a matched baseline
Sprouting	reproducible bifurcation into new path families	parameter sweep, stability analysis, and finite-size checks
Collective dependence	nonzero cross-response or nonproduct transition law	cross-coordinate generator term or statistical dependence test
Selection event	guard hit followed by a recorded jump	explicit reset law and outcome provenance
Stable basin	invariant or metastable set with quantified residence time	Lyapunov, spectral, large-deviation, or controlled numerical evidence
Irreversibility	failure of a stated reversal criterion	record/coarse-graining theorem or path-measure comparison

Table 3: Diagnostics required before attaching qualitative labels.

Several simple countermodels show why the labels are not inevitable. With $G_q = 0$, the barrier does not affect the drift. With constant speed and diffusion schedules, reduced capacity does not impose hesitation. With a reserve depending only on one coordinate, no channel branching need occur. With a stop policy, no post-guard state forms. With a bijective chart transition, the jump need not discard information.

13 Relation to established mathematical frameworks

13.1 Hybrid systems

The tuple of flow set, flow map, jump set, and jump map is standard in hybrid dynamics [5, 6]. The present model is not outside that class. Its distinctive bookkeeping is that the flow and jump sets are generated from a normalized multi-condition capacity record.

13.2 Viability and barrier methods

Viability theory asks whether trajectories can remain in a constrained set and studies viability kernels and exit tubes [2]. Control barrier functions provide derivative conditions for safety or forward invariance [1]. A capacity row can participate in such a certificate, but positivity at one instant does not establish invariance. The derivative condition must be checked for the supplied dynamics.

13.3 Mean-field and interacting particle systems

Fields built by summing kernels over all particles are standard interacting particle constructions. Their large- N limits can lead to McKean–Vlasov equations and propagation-of-chaos questions [4]. The MTT vocabulary does not remove the need for finite- N estimates, mean-field scaling, or convergence proofs.

13.4 What remains specifically useful here

The construction still contributes a useful research discipline:

1. begin with independently sourced, normalized admissibility rows;
2. preserve row identities rather than hiding them in one arbitrary scalar;
3. let the weakest row define a transparent guard;
4. declare the feedback and reset as separate model layers;
5. record whether the result stops, changes chart, or samples a new state; and
6. refuse downstream physical language until its required mathematical structure is present.

That composition can be valuable even though its component mathematics has close analogues.

14 Claim status and theorem ownership

This paper owns the following local results and definitions:

1. the typed capacity-gated hybrid record in Theorem 4.1;
2. the rowwise barrier construction tied to the MTT reserve record;
3. the barrier-feedback classification in Theorem 5.1;
4. the reset support contract in Theorem 6.1;
5. the no-selection result in Theorem 6.2;
6. the guarded hybrid well-posedness specialization in Theorem 7.1;
7. the shared-field cross-dependence criterion in Theorem 8.1; and
8. the complete executable and provenance contracts.

Standard SDE existence, strong Markov piecing-out, hybrid-system solution theory, viability theory, barrier certificates, mean-field limits, and quantum separability retain their literature ownership.

The following claims are *not* established:

- a universal MTT capacity evolution law;
- a geometry-selected relational kernel or reset kernel;
- inevitable guard hitting or inevitable basin formation;
- quantum entanglement, Bell correlations, or collapse;
- thermodynamic entropy production or an arrow of time;
- gravity, horizons, biological decision making, or consciousness;
- structural unpredictability beyond the chosen hybrid model; or
- superiority to established hybrid, control, or particle methods.

15 Completion and falsifiability contract

A proposed application is complete at the model level only if it supplies:

1. the exact upper state or action from which the rows are computed;
2. the fixed normalization, units, and hashes of all reserve rows;
3. the derivation or constitutive status of every kernel and coefficient;
4. the flow and jump sets, including tie and overlap conventions;
5. the reset map or kernel and its support proof;
6. a well-posedness or numerical-convergence certificate;
7. preregistered diagnostics and matched baselines;
8. uncertainty, finite-size, and discretization studies;
9. held-out validation data; and
10. a tier statement distinguishing toy model, reconstruction, fit, controlled numerical result, and selected prediction.

The model is falsified as a claimed application if, for example, its rows cannot be sourced, its reset leaves the declared admissible set, its reported behavior disappears under event or step refinement, its claimed locality uses a noncompact kernel without qualification, or its held-out predictions fail outside the construction data.

16 Conclusion

Capacity-gated projection dynamics can be made precise, but precision changes what may be claimed. The concrete construction is a hybrid stochastic particle model. Its relational fields provide classical coupling. Its logarithmic barrier provides gradient feedback. Its guard marks the boundary of one declared flow regime. Its reset law, not the guard, determines continuation.

That correction does not empty the idea. It turns it into a usable object. The model can now be simulated without silently crossing its domain, compared with established hybrid and mean-field methods, and promoted to a physical application only when an upstream source supplies its coefficients and reset. The central methodological lesson is simple: capacity can gate dynamics, but it does not write the dynamics for us.

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